

A new Net Zero doctrine

The Social Responsibility of Nations is to reduce debt

How to achieve climate and COVID recovery, by redirecting pension liability into net zero services



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Summary

Meeting the Paris agreement and development goals requires £5.2 trillion annual investment worldwide

Debt to build back better should be linked to and set against future pension obligations

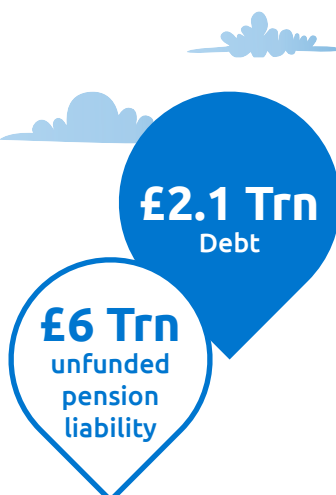
In the UK over £1 Trillion could be offset against unfunded pension obligations, helping to resolve the twin challenge of funding COVID recovery and net zero delivery by 2050

Applying sharing economy innovations can deliver “energy and mobility as a service” at scale for UK pensioners as a net zero universal basic service - a ‘low carbon right’

Radically rethinks how we meet our obligations to pensioners as societies age, whilst enabling pensioners to help deliver climate recovery



Net Zero pensions are an idea to help fund ‘build back better’ without increasing the long-term tax or debt burden. By re-imagining economic and pension policy, the UK can showcase a financial solution to the problem of finding the vast sums of capital needed to fund new infrastructure, new jobs and solving the climate emergency. This new net zero doctrine can be a centrepiece of COP26 and a platform for UK leadership.



£2.1 Trn
Debt

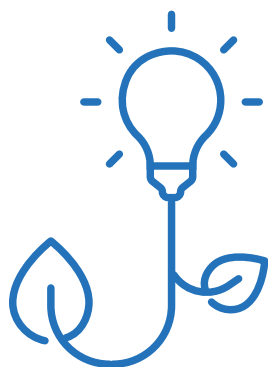
£6 Trn
unfunded
pension
liability

To meet the Paris climate agreement and G20 development goals, an estimated £5.2 trillion is needed annually to invest worldwide in the transition to low carbon energy and infrastructure, according to the OECD¹.

However, with UK debt exceeding **£2.1 Trillion** and with **£6 Trillion** in unfunded pension liabilities, we need a radical approach to avoid saturating the tax-payer and the next generation with additional tax and debt repayments. By combining the climate and pension challenges we can achieve a debt finance equivalent of BOGOF (buy one get one free) by funding climate infrastructure to deliver unfunded pension obligations as net-zero services.

Net Zero Pensions would link the twin challenges of population aging and pension shortfalls, with a solution that combines “As a Service” models from the sharing economy, artificial intelligence, renewable energy and electrification with the goal of enabling energy and mobility in retirement to be delivered in the form of net zero services.

Over one seventh of the UK’s £8 Trillion pension liability is spent on paying for energy and transport costs through retirement. This enables an approach to reduce financial and climate debt by redirecting a proportion of contracted pension cashflows. In return for a ‘Swap’ - giving up some of their pension income promise - future pensioners claim a new right in return; the provision of low-carbon “energy and mobility as a service” throughout their retirement. This ensures they never have to worry about the costs of keeping warm, or making a daily choice about affordability, whilst simultaneously contributing to climate recovery. More broadly, this new universal basic service is



analogous to the original idea for pensions: the historic notion of ‘corrody’ from the Middle Ages, where money invested in a Monastery earned a right to receive food and lodging for life. We already understand this idea in the UK, where citizens receive healthcare ‘as a service’ via the NHS, instead of receiving money to buy healthcare.

With increasing resource constraints, the sharing economy is already redefining the 21st century economy. The bulk of innovation has been seen in the private sector, such as the rise of Airbnb, which has found new ways to utilise housing and apartments as a more effective resource, through to the sharing economy solution of mobility as a service, from Uber and Lyft among others, as a more efficient way to extract value from under-used cars and vans. These same sharing economy concepts can be applied to the provision of public and energy services. Imagine a better world of low carbon transportation from the public sector, in which the government uses its collective spending power to direct infrastructure funding towards providing larger fleets of shared electric vehicles, or efficiencies in mega projects such as the £106bn HS2 if it can become low carbon at 8g per passenger km similar to what Eurostar achieved.

A key opportunity, however, lies in energy provision – by accelerating the provision of low carbon heating and electricity in homes we could achieve far lower carbon energy or even ‘no bills’ housing. Today, few homes use low carbon energy. All new and existing homes need to be low carbon to achieve net zero. While consumers are starting to move in this direction, selecting lower carbon energy providers, installing solar, batteries, LED lighting, and using smart meters, this gradual approach relies on consumer choices that will take far too long. The UK Government’s various energy White Papers similarly focus on short-term steps and policy nudges, rather than on the long-term outcome and legal requirements to actually deliver net zero by 2050. We need to move a lot faster than existing gradualist approaches.

Instead, the government could look at the substantial portion of the UK population where it plays the biggest role in the funding and provision of basic services: pensioners. In 2018, 18% of the total UK population was aged 65 years and over - compared with 10.8% in 1950 - and this is predicted to reach 24.8% in 2050. According to Ofgem, a typical existing household spends about 4-5% of their income on energy, with the poorest households spending as much as 12%. This is even starker for the elderly, where typically, over 14% of retirement income on an average basis is spent on energy bills and mobility through retirement. For those on low incomes this translates into fuel poverty, where over 10% of household income is spent on energy bills – often on inefficient buildings. According to a 2018 survey from Compare the Market, about 2.7 million elderly people in the UK ration their energy usage in winter, and almost a quarter of people over 65 had to reduce heating to make their energy payments. The Financial Times has even reported that the Queen was heading for fuel poverty given the rising costs of heating royal estates².

If you look at the bigger picture, it is bizarre that you have a convoluted system where you tax citizens to fund the Department of Work and Pensions’ growing £125bn annual managed expenditure and then use these commitments to pay state pensions in cash, from which a seventh of this income is then spent on paying for energy and transport.

Annually £7.6bn of green energy taxes are charged via Low Carbon Levies³ on energy bills to incentivise new low carbon infrastructure (currently the Levy Control Framework⁴ to pay for 'Contracts for Difference', 'Feed in Tariffs' and the Renewable Obligation). These do help incentivise low carbon, but the cost burden also falls on those least able to bear it, in the form of higher retail energy prices, which hurts the poor and elderly most.

There is a more direct approach the government could take. It could reallocate a proportion of unfunded state and defined benefit pension obligations that would have gone into paying energy and mobility costs. It could then invest and allocate that future cash-flow commitment today to fund debt to build low carbon infrastructure. The state could then become a dominant purchaser of this greener energy and deliver it back to future pensioners in the form of energy as a service, up to average consumption, rather than in income. Like the basic state pension, this could be basic, greener, state energy.

Of course, there will be a modest reduction in practical monthly pension income for some, but the benefits of long-term assured energy delivery, alongside improving the prospects of a greener future for younger generations, seems a small trade-off. To mitigate potential risks, where pensioners have low or no energy needs e.g. in care homes, they could trade their right for this low carbon energy service to future carbon markets for practical income. Governments could top up any potential contingent liabilities if there were a shortfall from renewables or technology failures. However, the certainty of assigning such significant cash-flows from unfunded pension liabilities enables practical funding of today's debt requirement for low carbon infrastructure and jobs.

To address the climate emergency and recover economically from COVID-19, the UK must use COP26 to adopt a new Net Zero doctrine⁵: "The new social responsibility of nations should be to reduce debt and mitigate climate breakdown by redirecting pension liability into net-zero services". We must build, serve and create jobs for a low carbon tomorrow, as a social imperative today.

UK's £8 Trillion pension liability

The UK, like most nations, faces significant unfunded liabilities across state, public and private pension scheme shortfalls. These exceed £6 Trillion in the UK and much more in Europe, where there are generous public sector commitments (which have caused recent public disorder as states try to raise pension eligibility ages). COVID-19 has also led to a significant increase in shortfalls, now exceeding £280bn across UK private sector schemes in the PPF7800 index⁶, and demand on the Pension Protection Fund.



The unfunded proportion represents a growing tax burden that will worsen as societies age. Over the next 25 years, the number of people over 70 is projected to rise by 52% compared to just 2% for those under 70⁷. Globally, this represents a £12 Trillion annual pension shortfall by 2050, according to G-30 research led by Adair Turner (Fixing the Pension Crisis)⁸. This financial gap cannot be fixed by taxing Millennials or Generation Z on top of the growing debt burden from the COVID recovery.

Both public and private schemes face the stark reality of unfunded liabilities, falling returns from traditional manufacturing industries, accelerating aging populations and falling tax takes, combined with manifesto promises on pensions, tax, and spending that look unrealistic and the threat of inflation.

However, significant savings can be made by merging these unfunded liabilities with debt funding of low carbon infrastructure, and through economies of scale and efficiencies gained from low carbon technologies that deliver future services at lower cost than conventional energy and avoid potential punitive carbon taxes.



Unlocking £1 Trillion from the pension system for Climate and COVID Recovery

The UK Budget should take a radical approach and create a new paradigm of pensions, climate finance and infrastructure recovery, instead of tax, debt and austerity.

This should set our agenda for COP26, piloted in the UK as a showcase for a Global Britain, Smart Energy Island approach to the future. This could provide an alternative to the multi-trillion US and EU Green Deals and 'bailouts' by creating a stable long term fiscal strategy that delivers a green recovery. This UK pilot could also show the new US government and the EU how to address fundamental social and debt issues on pensions and COVID and deliver on the Paris Accord.

Whilst the UK is already making significant progress in encouraging asset based pension funds, such as via Mark Carney's initiatives on ESG (Environmental, sustainable and governance) investing⁹, to de-risk and fund low carbon infrastructure, Net Zero Pensions is a 'whole system' approach addressing the larger problem of unfunded infrastructure and liabilities.

From a private level, some options on how it could work:

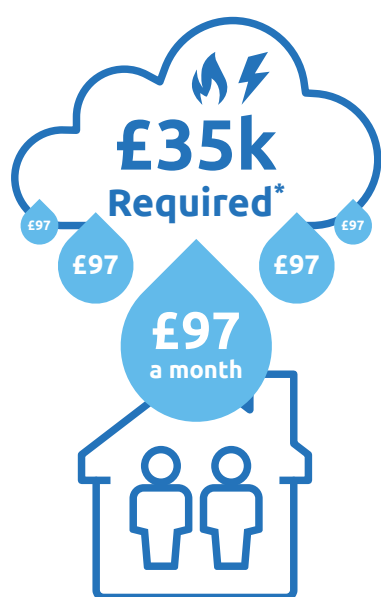
Tax relief on Pension drawdown and self-invested personal pension (SIPP) investments in low carbon home energy and mobility, as a potentially tax exempt drawdown – with tapering of relief if the asset is resold within a period.

The UK's original Green Deal and recent Green Home Grants are good initiatives but lack the financial scale to be impactful. The original Green Deal would have been more successful had households been empowered to draw down and co-invest in measures from their own pension and SIPP pots, with tax relief nudges, or to allocate from defined benefit scheme obligations, instead of borrowing money for high interest rates

As an example, each home requires an average £35,000 pension pot, just to buy an inflation linked annuity providing an income stream to cover paying for an average annual 'price-capped' energy bills. This is somewhat depressing, as Aegon estimate the average UK pension pot at £50k¹⁰, and private pensioners who have larger pots will often have larger homes with much higher energy bills.

Given this predicament, the significant benefit of investing in energy efficiency technology to reduce this life-long bill should be obvious to an informed retiree¹¹, such as through insulation, low energy heating and lighting or renewable energy to reduce annual bills and the need for pension income. For example, solar and batteries installed in homes help reduce lifetime energy bills and carbon, and also help society more widely by reducing overall electricity infrastructure costs. Deploying such technologies at scale and into the sharing economy has potential advantages on reducing sale, installation and procurement costs, as well as pooling technology and moving risks, and supporting delivery of national carbon targets.

Similarly, a pension pot of over £65,000 would be needed to fund the typical annual income needed to cover the estimated £2k annual running costs¹² of owning a petrol vehicle. Investing in an electric vehicle has significantly lower lifetime fuel costs thanks to electric charging, particularly when using smart charging to optimise the use of off-peak or low carbon electricity, and much lower maintenance costs¹³. By buying an electric vehicle battery you are in effect pre-paying your future fuel, and supporting the trajectory of electricity to be fossil free. It is also more fun to actually drive in your pension, rather than simply burning up an annuity income in fuel. With the potential of autonomous self-driving vehicles, the sharing economy could further enable the pooling of cars and pension pots or incomes. With massive energy capacity and trillions of pounds globally in electric vehicle batteries, shared batteries are also set to become critical for balancing grid infrastructure by storing and shifting solar and wind resources, potentially reducing the overall cost of low carbon energy for all.



Estimated pension pot required to pay life long energy bill

*Inflation linked to annuity at 65 for ONS average pensioner energy bill

Average "price cap" 2018-2020 £1-1.1k pa



Investments in housing and electric vehicles today can help the climate, economy and jobs, through a dramatic scale-up of 'truck-rolls'; installers working on homes for maintenance and improvements, as well as industries and manufacturing supply chains behind renewable and electric vehicle technologies.

Policies to encourage better release of private finance would raise awareness across financial advisors to help inform the wider opportunity with managed scheme freedoms and opt-ins from public and state pension schemes. It also presents improved focus for pension funds, in line with their ESG objectives to be better at advice and providing low carbon energy services. In simple terms, it's much better for pension companies to give you **LED lighting or solar panels** and work with households to share the risk of improvements by aggregating risk and purchase power, than simply holding these assets in order to pay out incomes to buy traditional energy supplies.

Scheme level - Reducing shortfalls through net zero 'bulk annuity' pension risk transfers

Defined benefit legislation should allow greater freedoms for scheme managers and members to opt into net zero services, such as via accelerating the pension risk transfer market with ESG targets for the proportion of scheme liabilities that relate to energy and mobility costs.

Defined benefit schemes face massive shortfalls. This could be reduced by enabling the option for scheme members to allocate a proportion of their pension into net zero energy and mobility services. Major companies, such as British Airways and BT, often trade their obligations as bulk annuities to funders better able to manage their assets for delivery. Given the £280bn missing across UK FTSE and PFF 7800 companies, there is a major opportunity for pension legislation to help schemes enable a partial opt-in to net zero pension delivery for energy and then mobility services. Empowering large scale delivery via the private and public sector of suitable services, from renewable energy, to electric vehicle sharing, to public transport infrastructure access.

Major energy, oil, automotive and communication corporations also face significant shortfalls in their pension liabilities, yet also often have emerging assets in renewables as well as delivery capabilities (retail, trading, installers) that could lend themselves to helping to deliver net zero services. Policies could encourage ESG targets, tax reliefs and favourable credits where these are partly used for internal scheme shortfalls by providing net zero services to former employees, in energy, mobility access and communication, instead of topping up scheme shortfalls from profits. This may allow balance sheet improvements by de-recognising some liabilities.

Considering the overall £280bn shortfall, a significant transfer or delivery of scheme obligations as services could crystallise significant balance sheet savings and improve FTSE performance, as well as drive a virtuous circle of internal investment in renewables, batteries, mobility technologies and jobs.

£280bn
private DB
shortfall

Nation level – Once in a generation reform to unfunded state and public pension liabilities

Unfunded defined benefit pension liabilities should be migrated to be matched with contracted net zero services. A new UK infrastructure bank should administer this debt funding for a green recovery from COVID. The Net Zero 2050 legal requirement should drive a long term fiscal policy and strategy to reduce debt.

Nations as well as municipal schemes, such as in the US which faces over \$1 trillion in unfunded liabilities for public¹⁴ city schemes, could simply mandate the provision of net zero energy and mobility services to achieve net-zero targets and reduce debt – with the direct byproduct of stimulating local infrastructure funding and jobs.

The UK should enable an opt-in to net zero services by encouraging a regular deduction from state pensions, and allow defined benefit scheme members to also make this choice. This swap should be emphasized in terms of its moral, economic and social benefits, particularly on the wider debt and climate debt impact. It may even be prudent to mandate it or, together with private and defined benefit scheme reliefs, require it in order to retain any triple lock mechanism on wider pension income paid as cash.

The public and private sector should be contracted to invest and provide net zero services. There is a role for overall Net zero debt to be administered by a new UK Investment and Infrastructure bank¹⁵. This together with ESG policies and recent TFCD (Taskforce for Climate related disclosures) and work of the Green Finance Institute can help direct investments into appropriate technologies. This can help make clearer the cleaner choices in energy, mobility, housing renovation and new-build, and extend more broadly into other potential net zero services such as in communication, sustainable food and other infrastructure.

An overall initial budget of £500bn in debt linked to pension liabilities should be allocated, increasing up to £1 Trillion in future, with a joined up strategy for low carbon delivery across government.

From a whole of government accounts (WGA)¹⁶ perspective, the UK faces £2,244bn in public sector pensions liabilities (£1,894bn un-funded). There is a further estimated £4,000bn in unfunded state pensions that are not recognised in the WGA as there is no contractual commitment to pay state pensions. Pensions are treated as a future policy choice and claim on potential future tax from a work-force which is in practice shrinking. However, by raising new debt today, a proportion of the unfunded public and state defined benefit liabilities becomes contractual, settled as net zero services and part of the WGA, so that the infrastructure debt is recognised against future tax cash-flow obligations.

There is also a moral imperative with the challenges of climate change. The generations that contributed to climate breakdown now have an opportunity to redress this by being part of the solution, directing part of their pension incomes to the trillions needed for a zero carbon world. For governments, reducing debt is a social imperative and through this mechanism climate recovery is funded as a by product.



How to spend it – A COP26 business plan for a better world

The UN Sustainable Development Goals set out the world's greatest business plan. It challenges nations to halve carbon in energy and mobility every decade¹⁷, to get rid of dependency on fossil fuels that damage our environment, air, livelihoods and raise the threat of survival of our planet¹⁸. Today, overall energy use and fuels in transport are responsible for two thirds of global emissions, equivalent to 35.3Gt of annual CO₂ equivalent emissions¹⁷.

To halve this requires strong leadership, pioneering technology and hundreds of billions of investment in shifting electricity systems to renewable energy generation and transport to low carbon, as well as improving energy efficiency in commercial and residential buildings. We need to rethink the whole system from national to home level.

The global growth in pensioner households can play an important role in providing a predictable customer base for services from infrastructure investments, underpinned by state-backed stable pension income. This could enable a national retrofit programme to improve household energy efficiency and reduce lifelong energy costs, and further encourage a electric vehicle ownership or 'usership' as shared low carbon transport. Delivering this great transformation and the economic opportunity of net-zero has the potential to create millions of jobs today for a low carbon tomorrow¹⁹.



Renewable energy generation:

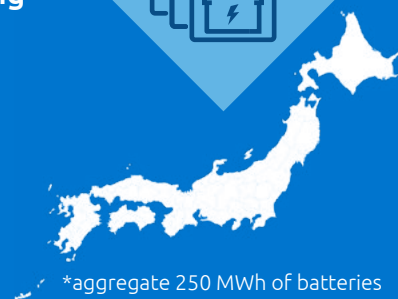
At the national level, we need to generate more electricity from solar and wind, either at grid scale or distributed across millions of households and rooftops. However, to achieve 100% clean electricity supply for retired and national households, we also need to invest in smart grid technology to leverage domestic energy storage and electric vehicle batteries that help store and shift renewable energy to peak times. New nuclear technology is also important, as it can deliver a low carbon base load, when wind and solar intermittent resources are unavailable.

Energy storage is particularly important to achieve net zero. Island nations, where the majority of electricity is generated locally, do not have the scale of interconnection to other countries to balance supply and demand as the energy system shifts to increasingly low carbon generation. Managing renewable supply and demand variations and keeping the whole system balanced requires flexibility technologies which can leverage energy storage or demand response to respond to price fluctuations and manage technical constraints on the energy system.

In South Australia, Tesla has demonstrated how a 194MWh/150MW energy storage facility at Hornsdale can help manage intermittent solar and household demand and network resilience.

Similarly, in Japan, Moixa has connected energy storage across 25,000 homes, providing an aggregate 250MWh battery, the world's largest example of a residential storage virtual power plant, which could be deployed to help support the grid as local flexibility markets and regulation evolve, and support Japan's migration away from nuclear power.

**25,000
homes**
providing an aggregate
250 MWh*



*aggregate 250 MWh of batteries

Low Carbon transportation:

Electrification of transport from buses to trains to cars significantly reduces lifelong carbon emissions. This requires significant investment in new battery technologies, gigafactories for production, electric charging infrastructure and control of domestic charging, as well as the production of electric vehicle fleets and their batteries.

There are considerable advantages to greater access to shared vehicles, which have economic, waste and utilisation benefits over owned vehicles - which are often unused 95% of the time. Indeed, we have likely passed peak vehicle ownership. The rise of shared and, in future, autonomous vehicles will also help increase access for aging populations who become less able to drive themselves and younger generations who increasingly choose not to learn to drive.

Various options to access different types of transportation can be approached as a universal basic service. For example, bus access in the UK is already free at pensionable age. This could be extended for tiered access to trains, electric vehicle sharing and driving services, as part of a package of low carbon services linked to an exchange of a portion of the pension income promise. Automation itself is expected to transform the entire notion of car ownership into part ownership with access rights to shared vehicle resources. The sharing economy will be prevalent in the net zero world of 2050.

In the UK there are over 200,000 electric vehicles, with around 8GWh of batteries which is as large as hydro-power plants in the UK. This is expected to rise to nearly 1TWh with full electrification of UK vehicle fleets accelerated by 2030 bans on petrol and diesel car sales.

Worldwide there are forecast to be 250m electric vehicles by 2030, or over 10TWh of batteries on the move, with asset costs into trillions of pounds. These can also be harnessed to create major virtual power plants, enabling more efficient renewable energy supply and lower overall cost.

The growth in electric vehicles will add significant demand to electricity grids but can also be managed, or smart-charged, to support the grid and the availability of renewable energy, and be orchestrated to help avoid new network upgrade costs. Artificial Intelligence can help learn and optimise smart charging and coordinate distributed energy resources such as batteries and the sharing of spare capacity to the grid, delivering new flexibility services and better economic returns. By harnessing millions of batteries, every home, business and electric car can support a cleaner energy system.

Energy Efficient Households:

Investing net zero infrastructure funding in upgrading the nations housing stock to be more energy efficient, and in new net zero house building, has the potential to create immediate jobs at scale. Worldwide, over 400m new homes are expected to be built in the next decade²⁰. It is essential these are low carbon and built with the latest clean technologies and standards.

Solar roofs, smart batteries, heat pumps, insulation and passive house standards, LED lighting, together with smart controls, for new build or retrofit, can significantly reduce the ecological footprint. In particular, investing now can reduce lifelong energy costs and therefore the need for income to pay for energy. This can be predictable for pensioner homes and new builds and enable energy as a service models, particularly when linked to house purchase, rent or pension cash flows paying back technology costs over a term.

Smart Charging the future

As the world moves away from dependency on global supply chains and interconnection, the role of financing and supporting local energy resources will become essential to deliver a resilient system²¹. Linking today's infrastructure spending to provide universal basic low carbon services, for a predictable demand for energy and mobility, can help fulfill state backed pension obligations and reduce the overall future tax and debt burden.

The world needs to invest trillions of pounds on the clean transition and to economically recover from COVID-19. Clean energy infrastructure provides synergy²² with investments that create jobs, to rebuild and deliver a new low carbon tomorrow. But the key is financing through Net Zero Pensions to redirect liabilities systemic across nations to afford and achieve this common goal.

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